

# Deba Priyo Guha

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## Summary

A passionate research enthusiast and machine learning practitioner specializing in astronomical data science and deep learning applications. Strong foundation in competitive programming, problem-solving, and interdisciplinary research bridging astrophysics and artificial intelligence. Demonstrated leadership in academic organizations and commitment to STEM education through olympiad participation, tutoring, and mentorship. Focused on developing interpretable AI frameworks for scientific discovery, particularly in radio astronomy and astronomical transient classification. Experienced in full research pipeline from data preprocessing to manuscript preparation, with expertise in handling large-scale astronomical datasets and multimodal AI systems. Active contributor to competitive programming communities and AI training platforms. Selected for Bangladesh National Team for International Olympiad on Astronomy & Astrophysics (IOAA 2017).

## Research Interests

**Primary Areas:** Machine Learning and Deep Learning in Astrophysics, Radio Galaxies, Astronomical Data Science, Computer Vision for Scientific Imaging, Medical Image Analysis, Multimodal AI

**Specific Focus:** Interpretable AI frameworks for scientific discovery, Automated astronomical classification systems, Radio Active Galactic Nuclei (AGN) morphological classification, Astronomical transient detection and classification, Lunar hyper-spectral mineralogy mapping, Visual Large Language Models for multi-wavelength galaxy characterization, Transformer-based architectures for scientific applications, Generative models for data augmentation in scientific imaging, Time-domain astrophysics, Physics-informed deep learning

## Education

### University of Dhaka

*Master of Science in Theoretical Physics*

Dhaka University Campus, Dhaka-1000, Bangladesh

*March 2026–Present*

### Current Coursework:

- TPG501 | Introduction to Condensed Matter Physics | 4 Cr.
- TPG525 | Non-equilibrium Statistical Mechanics | 4 Cr.
- TPG550 | General Relativity and Cosmology | 3 Cr.
- TPG555 | Relativistic Quantum Mechanics and QED | 4 Cr.
- TPG565 | Quantum Mechanics-1 | 3 Cr.
- TPG575 | Introduction to The Standard Model-1 | 3 Cr.
- TPG598 | General Oral Examination | 4 Cr.
- TPG599 | Thesis | 6 Cr.
- TPG541 | Group Theory and Representation Theory | 3 Cr.

### Rajshahi University of Engineering & Technology (RUET)

*Bachelor of Science in Engineering*

Department of Electrical & Computer Engineering (ECE)

Kazla, Rajshahi-6204, Bangladesh

CGPA: 2.88/4.00

Graduation Date: 9 September 2024

[🌐 https://ruet.ac.bd/](https://ruet.ac.bd/)

*Feb 2019–Sept 2024*

**Thesis:** "PIE-ADA: Physics-Informed Ensemble with Adaptive Data Augmentation for Photometric Transient Classification"

Supervisor: Fariya Tabassum (Assistant Professor & Head, ECE Department)

**Relevant Coursework:** Circuits and Systems I Circuits and Systems I Sessional Computer Programming Computer Programming Sessional Calculus and Co-ordinate Geometry Optics and Modern Physics Optics and Modern Physics Sessional Technical English Technical English Sessional Introduction to Computer System Circuits and Systems II Circuits and Systems II Sessional

Object Oriented Programming Object Oriented Programming Sessional Analog Electronic Circuits-I Analog Electronic Circuits-I Sessional Economics, Government Sociology Differential Equations Electrical Shop Practice Data Structure & Algorithms Data Structure & Algorithms Sessional Electrical Machines-I Electrical Machines-I Sessional Digital Techniques Digital Techniques Sessional Inorganic and Physical Chemistry Inorganic and Physical Chemistry Sessional Vector Analysis & Linear Algebra Software Development Project-I Analog Electronic Circuits-II Analog Electronic Circuits-II Sessional Numerical Methods Numerical Methods Sessional Data Base Systems Data Base Systems Sessional Complex Variable, Statistics & Transform Methods Industrial Management & Accountancy Software Development Project-II Electrical Machines-II Electrical Machines-II Sessional Power System Microprocessors & Assembly Language Microprocessors & Assembly Language Sessional Software Engineering & Information System Design Software Engineering & Information System Design Sessional Computer Architecture and Design Electronic Shop Practice Industrial Electronics Industrial Electronics Sessional Communication Engineering Communication Engineering Sessional Peripheral and Interfacings Peripheral and Interfacings Sessional Computer Aided Instrumentations Computer Aided Instrumentations Sessional Operating System Operating System Sessional Power Station, Switchgear & Protection Electromagnetic Fields & Waves Digital Signal Processing Digital Signal Processing Sessional Computer Networks Computer Networks Sessional Power System Operation & Control Power System Operation & Control Sessional Industrial Training Seminar VLSI Design Digital Image Processing Digital Image Processing Sessional Control Systems Control Systems Sessional Neural Networks & Fuzzy Systems Neural Networks & Fuzzy Systems Sessional Fiber Optic Communication Fiber Optic Communication Sessional Thesis Project-II

### **Govt. City College Chattogram**

2016–2018

*Higher Secondary School Certificate (HSC)*

7 Ice Factory Road, Chattogram-4000, Bangladesh

GPA: 4.83/5.00

Group: Science

🌐 <https://gcc.edu.bd/>

### **Nasirabad Govt. High School**

2010–2016

*Secondary School Certificate (SSC)*

CDA Avenue, East Nasirabad, Chattogram, Bangladesh

GPA: 5.00/5.00

Group: Science

🌐 <https://nghs.tsmts.com/>

### **St. Xavier's School, Chattogram**

2003–2009

*Primary Education*

Chattogram, Bangladesh

🌐 <https://www.sxsctg.edu.bd/>

### **Language Proficiency:**

**IELTS Academic:** Overall Band Score 7.0 (Listening: 8.0, Reading: 7.5, Writing: 6.5, Speaking: 6.0)

Test Date: 22 September 2025 | Test Center: British Council, Bangladesh

## **Research Experience**

### **Postbaccalaureate Research Intern**

Oct 2025–Present

*Center for Astronomy, Space Science & Astrophysics (CASSA)*

Independent University, Bangladesh (IUB)

Supervisor: Dr. Khan Muhammad Bin Asad

🌐 <https://cassa.site/>

- Contributing to the development of RGC v2.0 (Radha Gobinda Chandra) Python package for automated morphological classification of radio Active Galactic Nuclei (AGN)
- Developing hierarchical radio AGN classifier for bent morphologies (Wide-Angle Tail, Narrow-Angle Tail) using 20,000+ LOFAR/LoTSS and VLA/FIRST images
- Implementing semi-supervised Convolutional Neural Network (CNN) models for classification of bent radio AGN morphologies
- Implementing semi-supervised learning frameworks and spurious source filtering algorithms for radio astronomical surveys
- Developing Visual Large Language Model (LLM) pipeline for multi-wavelength counterpart galaxy characterization and automated description generation
- Building multimodal AI system ingesting galaxy images across optical, IR, radio, and X-ray bands for automated astrophysical characterization
- Reviewing and revising RGC v1 manuscript in response to journal committee; assisting in dataset re-preparation for resubmission
- Gaining full research pipeline experience from data acquisition, preprocessing, model development, validation, to manuscript preparation
- Working with high-performance computing resources for large-scale astronomical data processing

- Collaborating with international astronomy research teams

### Postbaccalaureate Research Assistant

*Feb 2026–Present*

*Center for Astronomy, Space Science & Astrophysics (CASSA)*

Independent University, Bangladesh (IUB)

Supervisor: Dr. Syed Ashraf Uddin (Associate Professor, Core Member, CASSA, IUB)

✉ [uddinsa@iub.edu.bd](mailto:uddinsa@iub.edu.bd)

- Re-analysing host galaxy stellar population ages of Type Ia supernovae from Rose et al. (2019) using Nautilus neural-network nested sampling in place of emcee MCMC, with bit-identical likelihood and priors
- Investigating systematic biases in SN Ia standardisation arising from sampler choice; identified  $+0.776 \pm 0.289$  Gyr offset ( $2.69\sigma$ ) between Nautilus and MCMC ages across  $N=98$  host galaxies
- Developing Bayesian SED-fitting pipeline (BAGPIPES + Nautilus) on HPC cluster (timaeus server) with automated SLURM job management
- Applying Uddin et al. (2017) intrinsic scatter formalism to quantify host galaxy age step in SN Ia Hubble residuals
- Investigating host galaxy age correlations with SN Ia light-curve parameters and Hubble residuals across young/old host subsamples
- Writing first-author journal paper (target: Astronomy & Astrophysics) on sampler systematics in SN Ia host galaxy analysis

### Research Collaborator

*2024–Present*

*Young Learners' Research Lab*

Department of Computer Science & Engineering, RUET

Supervisor: Md. Azmain Yakin Srizon (Assistant Professor)

- Leading multiple machine learning research projects focusing on astronomical transient classification and lunar hyperspectral analysis
- Developing LunaXNet framework for mineralogical segmentation, quantification, and interpretation from Chandrayaan-1 M3 lunar hyperspectral data
- Developing hybrid CNN-Transformer framework for lunar mineralogical segmentation from Chandrayaan-1 M3 hyperspectral data (85 spectral bands, 430–3000 nm)
- Implementing Grad-CAM based interpretability analysis for olivine, pyroxene, and plagioclase identification on lunar surface
- Co-authoring research projects on medical image classification using Generative Adversarial Networks (GANs) with interpretable frameworks
- Contributing to transformer-based Natural Language Processing (NLP) research for spam detection using BERT and GPT-2 architectures
- Mentoring junior students in machine learning concepts and research methodologies
- Conducting literature reviews and staying current with latest developments in ML/DL research

### Undergraduate Thesis Researcher

*Jan 2024–Sept 2024*

*Department of Electrical & Computer Engineering, RUET*

Supervisor: Fariya Tabassum (Assistant Professor & Head, ECE Department)

Credit Hours: 4.5

Thesis Title: *"PIE-ADA: Physics-Informed Ensemble with Adaptive Data Augmentation for Photometric Transient Classification"*

- Developed PIE-ADA (Physics-Informed Ensemble with Adaptive Data Augmentation) for photometric transient classification targeting LSST survey
- Developed comprehensive machine learning framework for 14-class astronomical transient classification using the PLAsTiCC (Photometric LSST Astronomical Time-Series Classification Challenge) dataset containing 7,848 original samples augmented to 8,148
- Implemented astrophysically constrained synthetic data generation: time dilation, interstellar extinction, correlated noise injection
- Extracted 271 multi-scale features from six photometric passbands
- Implemented and compared multiple ML algorithms including LightGBM, XGBoost, Random Forest, Extra Trees, Neural Networks, Gradient Boosting, Support Vector Machines, Logistic Regression, LSTM networks, and CNN architectures
- Achieved weighted logarithmic loss of 0.5767 ( $\pm 0.0089$ ) and 80.36% accuracy using 5-fold cross-validation
- Achieved 24–33% improvement over baseline algorithms (32–48% improvement in some metrics)
- Addressed severe class imbalance issues (some classes  $<1\%$  of data) using SMOTE (Synthetic Minority Over-sampling Technique), ADASYN, class weighting strategies, and Physics-Informed Adaptive Data Augmentation
- Performed extensive feature engineering on time-series photometric data including magnitude, flux, and observation metadata
- Classified supernovae types (Type Ia, Ib/c, II), kilonovae, superluminous supernovae, pair-instability supernovae, Active Galactic Nuclei (AGN), tidal disruption events, RR Lyrae, M-dwarf flares, eclipsing binaries, and other astronomical transients
- Demonstrated successful application of data-driven techniques to complex astrophysical classification challenges
- Conducted comprehensive error analysis and model interpretability studies
- Implemented cross-validation and temporal validation strategies to prevent data leakage
- PIE-ADA provides a computationally lightweight and interpretable method of transient classification at a speed that can perform real-time tasks

- Seamlessly blended computational machine learning skills with astrophysical domain knowledge

## Publications & Manuscripts

### Published

1. **D. P. Guha**, F. Tabassum, “PIE-ADA: Physics-Informed Ensemble with Adaptive Data Augmentation for Photometric Transient Classification,”  
*2026 IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence & Networking (QPAIN 2026)*, IEEE, 2026

**DOI:** [10.1109/QPAIN69676.2026.11546175](https://doi.org/10.1109/QPAIN69676.2026.11546175)

**Abstract:** The upcoming Large Synoptic Survey Telescope (LSST) is expected to observe approximately 10 million astronomical transient events per night, creating an urgent need for automated classification. A key challenge is the extreme class imbalance in transient datasets, where rare event types represent less than 1% of all observations. This paper presents PIE-ADA (Physics-Informed Ensemble with Adaptive Data Augmentation), a framework that generates physically realistic synthetic light curves for underrepresented classes using astrophysically motivated transformations. PIE-ADA applies four augmentation operations, namely correlated noise injection, cosmological time dilation, wavelength-dependent dust extinction, and observation phase shifting, while enforcing physical constraints to prevent unrealistic samples. We extract 271 multi-scale features from six photometric passbands covering statistical, temporal, peak, color, and frequency-domain properties. Evaluated on the PLAsTiCC dataset (7,848 original objects augmented to 8,148 across 14 classes), five classifiers were compared using stratified 5-fold cross-validation. LightGBM achieved the best performance with a weighted log loss of 0.5763 ( $\pm 0.0083$ ) and 80.33% accuracy, improving over Random Forest, Extra Trees, and Neural Network baselines by 24–49% in log loss. The framework is computationally efficient, completing the full pipeline in under 37 minutes and classifying individual objects in less than 0.05 seconds, making it suitable for real-time LSST alert processing.

**Status:** Published — IEEE Xplore

### In Preparation (First Author)

2. **D. P. Guha**, M. A. Y. Srizon, et al., “LunaXNet: A Novel Pre-Post-AI Framework for Mineralogical Segmentation, Quantification, and Interpretation from Lunar Hyperspectral Data,”

*Target Journal: IEEE Transactions on Geoscience and Remote Sensing*

**Abstract:** We introduce LunaXNet, an end-to-end deep learning framework for automated lunar mineralogy mapping from Chandrayaan-1 Moon Mineralogy Mapper (M3) hyperspectral data. The framework integrates three key components: (1) Pre-AI preprocessing pipeline handling atmospheric correction, photometric normalization, and spectral noise reduction; (2) LunaXNet architecture—a novel hybrid CNN-Transformer model for semantic segmentation of mineral species (olivine, pyroxene, plagioclase, ilmenite, spinel) at sub-kilometer resolution; (3) Post-AI interpretation module providing quantitative mineral abundance maps, confidence metrics, and geological context through integration with lunar topography and crater age databases. We train and validate on M3 Level 2 data covering diverse lunar terrains (mare basalts, highland anorthosites, impact ejecta). LunaXNet achieves 92%+ pixel-wise accuracy, outperforming traditional spectral unmixing and existing CNN baselines. Grad-CAM visualization confirms the model learns physically meaningful spectral features corresponding to crystal field absorptions. Our framework enables rapid, consistent mineralogical analysis of the 6 TB M3 archive, supporting lunar science and resource prospecting. We release code, trained models, and processed mineral maps to facilitate reproducible lunar remote sensing research.

**Status:** Model architecture finalized, experiments 60% complete, manuscript in early draft stage

### Under Review / Submitted (Co-author)

3. M. S. Hossain, M. S. H. Shahal, A. Khan, K. M. B. Asad, P. Saikia, M. O. B. Jihad, **D. P. Guha**, et al., “RGC: A Radio AGN Classifier Based on Deep Learning. I. A Semi-Supervised Model for the VLA Images of Bent Radio AGNs,”

*Submitted to: Astronomy & Astrophysics*

**arXiv:** [arXiv:2510.22190](https://arxiv.org/abs/2510.22190)

**My Contribution:** Vision-LLM pipeline development, multi-wavelength data integration, dataset preparation for resubmission, responding to journal review

**Status:** Under review at Astronomy & Astrophysics (A&A), March 2026

### In Preparation (First Author, Co-supervised)

4. **D. P. Guha**, S. A. Uddin, “Probing Systematic Biases in Type Ia Supernova Standardisation: Host Galaxy Dependencies, Local Environment Effects, and Cross-Calibration of Light-Curve Fitters,”

## In Preparation (Co-author)

5. M. S. H. Shahal, A. Khan, **D. P. Guha**, K. M. B. Asad, et al., “RGC v2.0: Hierarchical Multi-Class Classification of Radio AGN Morphologies,”  
*Target Journal: Astronomy & Astrophysics*  
**Abstract:** We present RGC v2.0, an extended version of the Radha Gobinda Chandra radio galaxy classifier, featuring hierarchical multi-class classification of radio AGN morphologies. The framework processes images from LOFAR/LoTSS (144 MHz) and VLA/FIRST (1.4 GHz) surveys, implementing semi-supervised CNN architecture for classification of Fanaroff-Riley Type I (FRI), Type II (FRII), Wide-Angle Tail (WAT), Narrow-Angle Tail (NAT), Compact, and Hybrid morphologies. We leverage limited labeled data ( 500 expert classifications) combined with large unlabeled candidate pools using semi-supervised learning techniques.  
**My Contribution:** Semi-supervised learning implementation, bent morphology classification (WAT/NAT), data preprocessing pipeline development, model training and validation
6. M. S. Hossain, et al., **D. P. Guha**, K. M. B. Asad, et al., “Vision-Language Models for Multi-Wavelength Galaxy Characterization,”  
*Target Journal: The Astrophysical Journal Supplement Series*  
**Abstract:** We present a multimodal AI system leveraging Vision-Language Models for automated characterization of galaxies across multiple wavelengths. The system ingests galaxy images from optical (Pan-STARRS, SDSS), infrared (WISE), radio (LOFAR, VLA), and X-ray surveys to generate comprehensive astrophysical descriptions and classifications.  
**My Contribution:** Vision LLM pipeline development, multi-wavelength data integration, automated description generation system
7. T. Sabira, **D. P. Guha**, M. A. Y. Srizon, et al., “GAN-Enhanced Gastrointestinal Disease Classification with Dual-Stage Preprocessing and Interpretable Post-Processing Framework,”  
*Target Journal: Medical Image Analysis (Elsevier)*  
**Abstract:** Gastrointestinal disease diagnosis from endoscopic imagery faces challenges of class imbalance, limited training data, and interpretability requirements for clinical adoption. We propose a comprehensive pipeline integrating: (1) Dual-stage preprocessing: color normalization via histogram matching plus adaptive histogram equalization for contrast enhancement; (2) GAN-based data augmentation: StyleGAN2-ADA generates synthetic endoscopic images for minority classes (esophagitis, polyps), validated by Fréchet Inception Distance and expert radiologist review; (3) Multi-architecture ensemble: EfficientNetV2, ResNet152, and Vision Transformer (ViT) classifiers trained on augmented Kvasir-SEG and Kvasir-V2 datasets; (4) Interpretable post-processing: Grad-CAM and SHAP analyses highlight diagnostically relevant image regions, validated against clinician annotations. Our framework achieves 96.2% accuracy across 8 disease classes, with balanced performance on rare conditions. Ablation studies quantify GAN augmentation’s 7.3% improvement over baseline. Grad-CAM heatmaps align with expert-identified pathological features (ulceration, vascular patterns, polyp margins) in 91% of cases. We discuss computational requirements, failure modes on ambiguous cases, and clinical deployment considerations including inference speed and confidence calibration.  
**My Contribution:** GAN implementation and training, synthetic data quality assessment, ensemble model integration, Grad-CAM visualization, literature review, manuscript co-
8. S. Rezwana, **D. P. Guha**, “Physics-Inspired Multimodal Emotion Recognition for Telemedicine Triage,”  
*Target: 2026 IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence & Networking (QPAIN 2026)*  
**Abstract:** We present a physics-inspired multimodal emotion recognition framework for telemedicine triage applications. Using the MELD (Multimodal EmotionLines Dataset), we map conversational emotions to clinical risk levels (high/mid/low) for remote healthcare prioritization. Our approach introduces two key contributions: (1) Telehealth Acoustic Augmentation Bank (TAAB)—a physics-inspired audio augmentation pipeline simulating realistic telehealth conditions including reverberation, ambient noise, bandwidth limitations, and speech rate variations; (2) A light physics-aware loss function incorporating temporal and spectral smoothness constraints inspired by acoustic wave propagation principles. The multimodal architecture fuses BERT-based text embeddings with physics-aware CNN audio features. Experiments demonstrate that TAAB augmentation improves audio-only baseline performance, while the physics-aware loss provides additional regularization benefits. The fusion model (BERT + TAAB + physics-loss CNN) achieves superior performance on high-risk emotion detection (sadness, fear, anger, disgust), critical for telemedicine triage prioritization. We provide Grad-CAM visualizations and token importance analysis for model interpretability.  
**My Contribution:** TAAB (Telehealth Acoustic Augmentation Bank) design and implementation, physics-aware loss function formulation, ablation studies, related work on physics-informed neural networks and audio augmentation, discussion section co-writing  
**Status:** Implementation in progress, experiments 30% complete

# Selected Research Projects & Technical Work

## **RGC v2.0: Radha Gobinda Chandra—A Radio Galaxy Classifier Package**

*Oct 2025–Present*

*Center for Astronomy, Space Science & Astrophysics (CASSA), IUB*

Supervisor: Dr. Khan M. B. Asad

🌐 <https://cassa.site/entry/rgc/>

- Developing open-source Python package for automated morphological classification of radio Active Galactic Nuclei
- Processing and analyzing 20,000 radio galaxy images from LOFAR LoTSS (144 MHz) and VLA FIRST (1.4 GHz) surveys
- Implementing multi-class classification pipeline: Fanaroff-Riley Type I (FRI), Type II (FRII), Wide-Angle Tail (WAT), Narrow-Angle Tail (NAT), Compact, and Hybrid morphologies
- Building semi-supervised CNN architecture leveraging limited labeled data (500 expert classifications) and large unlabeled candidate pool
- Developing Visual Large Language Model (LLM) component for automated multi-wavelength galaxy characterization combining radio, optical (Pan-STARRS), and infrared (WISE) imagery
- Creating preprocessing pipeline: source extraction, image cutout generation, resolution matching, noise filtering, and normalization
- Implementing cross-survey matching and multi-frequency analysis tools
- Designing user-friendly API for community adoption in radio astronomy workflows
- Utilizing HPC resources for large-scale inference across full survey footprints
- Contributing to scientific discovery of rare morphological classes and hybrid sources

## **PIE-ADA: Astronomical Transient Classification Using Machine Learning**

*Jan 2024–Sept 2024*

*Department of Electrical & Computer Engineering, RUET*

Supervisor: Fariya Tabassum | Credit: 4.5 | *Undergraduate Thesis Project*

- Developed PIE-ADA (Physics-Informed Ensemble with Adaptive Data Augmentation) for photometric transient classification
- Developed comprehensive ML framework for 14-class astronomical transient classification using PLAsTiCC dataset (7,848 original samples, 1.4M+ time-series observations)
- Classified: Type Ia supernovae (various subtypes), Type Ibc supernovae, Type II supernovae, kilonovae, superluminous supernovae, pair-instability supernovae, tidal disruption events, AGN, RR Lyrae, M-dwarf flares, eclipsing binaries
- Implemented astrophysically constrained synthetic data generation: time dilation, interstellar extinction, correlated noise injection
- Extracted 271 multi-scale features from six photometric passbands
- Implemented and systematically compared algorithms: LightGBM, XGBoost, Random Forest, Extra Trees, Neural Networks, Gradient Boosting, AdaBoost, Support Vector Machines (SVM), Logistic Regression
- Explored deep learning approaches: Long Short-Term Memory (LSTM) networks for time-series modeling, Convolutional Neural Networks (CNN) for feature extraction
- Performed extensive feature engineering: statistical features (mean, median, std, skewness, kurtosis), peak magnitudes, rise/fall times, color indices, observation cadence patterns
- Addressed severe class imbalance (some classes <0.1% of data) using SMOTE, ADASYN, class weighting, ensemble under-sampling, and physics-informed augmentation
- Achieved weighted log-loss of 0.5767 ( $\pm 0.0089$ ) and 80.36% accuracy with LightGBM ensemble
- Achieved 24–33% improvement over baseline algorithms
- Conducted detailed error analysis: confusion matrices, per-class precision/recall, misclassification patterns
- Analyzed feature importance: identified key discriminative features for each transient class
- Implemented cross-validation and temporal validation strategies to prevent data leakage
- Compared computational efficiency and scalability across algorithms for large-scale survey applications

## **LunaXNet: Lunar Mineralogy Mapping from Hyperspectral Data**

*2024–Present*

*Young Learners' Research Lab, Department of CSE, RUET*

Supervisor: Md. Azmain Yakin Srizon

- Developing deep learning framework for semantic segmentation and quantitative analysis of lunar minerals using Chandrayaan-1 M3 (Moon Mineralogy Mapper) hyperspectral data
- Processing high-dimensional spectral cubes: 85 continuous bands covering 430–3000 nm wavelength range at 20–40 nm spectral resolution
- Targeting mineral species: olivine, pyroxene (low-Ca and high-Ca), plagioclase feldspar, ilmenite, spinel, glass
- Building preprocessing pipeline: photometric correction (removing illumination effects), thermal emission removal, atmospheric correction, spectral smoothing, continuum removal
- Designing LunaXNet architecture: hybrid CNN-Transformer model combining spatial context (CNNs) with spectral attention mechanisms (Transformers)
- Implementing spectral feature extraction: identifying crystal field absorptions (1000 nm, 2000 nm bands), charge transfer features, spectral slopes
- Creating ground-truth labels from Apollo sample return sites, spectral libraries (RELAB, USGS), and previous mineralogical studies

- Developing post-AI interpretation: mineral abundance quantification, confidence estimation, geological context integration
- Performing Grad-CAM analysis to verify model focuses on physically meaningful spectral features
- Targeting specific lunar regions: Orientale Basin, South Pole-Aitken Basin, Aristarchus Plateau, Apollo landing sites
- Processing multi-TB dataset covering significant lunar surface area

### **Vision-Language Models for Multi-Wavelength Galaxy Characterization**

2025–Present

*Center for Astronomy, Space Science & Astrophysics (CASSA), IUB*

Supervisor: Dr. Khan M. B. Asad

- Building multimodal AI system for automated galaxy characterization across multiple wavelengths
- Integrating galaxy images from optical (Pan-STARRS, SDSS, Legacy Survey), infrared (WISE), radio (LOFAR, VLA), and X-ray surveys
- Implementing Vision-Language Model architectures (CLIP, BLIP) for astronomical image understanding
- Developing automated description generation system for galaxy morphology, spectral properties, and astrophysical characteristics
- Creating cross-matching pipelines for multi-wavelength source identification
- Designing evaluation metrics for astronomical description quality

### **GAN-Enhanced Medical Image Classification**

2024–Present

*Young Learners' Research Lab, Department of CSE, RUET*

Role: Co-author | Supervisor: Md. Azmain Yakin Srizon

- Addressing class imbalance in gastrointestinal disease classification using Generative Adversarial Networks
- Datasets: Kvasir-SEG (1000 polyp images with segmentation masks), Kvasir-V2 (8000 GI images across 8 classes)
- Disease classes: normal, esophagitis, polyps, ulcerative colitis, dyed-lifted polyps, dyed-resection margins
- Implemented StyleGAN2-ADA for generating synthetic endoscopic images of minority classes
- Validated synthetic images: Fréchet Inception Distance (FID) scores, visual inspection by domain experts
- Built dual-stage preprocessing: color normalization via histogram matching, adaptive histogram equalization (CLAHE)
- Trained multi-architecture ensemble: EfficientNetV2-L, ResNet152, Vision Transformer (ViT-B/16)
- Achieved 96.2% classification accuracy with balanced performance across all classes
- Implemented interpretability frameworks: Grad-CAM heatmaps highlighting diagnostically relevant regions
- Validated Grad-CAM outputs against expert annotations of pathological features
- Conducted ablation studies quantifying contribution of GAN augmentation, preprocessing stages, and ensemble components
- Analyzed failure modes: performance on ambiguous cases, inter-observer variability effects

### **Car Counter: Real-Time Vehicle Monitoring System**

2022

*ECE IoT Lab, RUET*

Supervisor: Tasnim Binte Shawkat | Credit: 1.50

- Designed and implemented IoT system for real-time vehicle detection, counting, and traffic density estimation
- Hardware: Raspberry Pi 4 Model B, Pi Camera Module v2, power supply, mounting hardware
- Software stack: Python 3.7, OpenCV for computer vision, NumPy for numerical operations
- Implemented vehicle detection: background subtraction (MOG2), blob detection, contour analysis
- Built tracking algorithm: centroid tracking for unique vehicle identification across frames
- Developed counting logic: virtual detection lines, direction-based counting (in/out)
- Optimized for real-time performance: frame resizing, multi-threading, GPU acceleration
- Created web dashboard: Flask backend, real-time count display, historical data visualization
- Tested under various conditions: different lighting, weather, traffic densities, vehicle types
- Achieved 90%+ counting accuracy in moderate traffic conditions
- Enhanced skills: embedded systems programming, real-time image processing, hardware-software integration

### **knackbook: Social Media Platform for Talent Showcasing**

2021

*Department of ECE, RUET*

Supervisor: Rakibul Hassan | Credit: 0.75

- Designed and developed full-stack social networking platform focused on creative talent sharing
- Frontend: HTML5, CSS3, Bootstrap 4 for responsive design, JavaScript/jQuery for interactivity
- Backend: PHP for server-side logic, session management, authentication
- Database: MySQL for relational data storage, normalized schema design
- Core features: user registration/login with validation, profile creation and customization, photo/video uploads with file handling
- Social features: post creation with media attachments, commenting system, real-time messaging between users, like/reaction system
- Implemented security: password hashing (bcrypt), SQL injection prevention (prepared statements), XSS protection, CSRF tokens
- Built newsfeed algorithm: chronological and relevance-based post ordering
- Developed search functionality: user search, hashtag support, content filtering
- Created notification system: activity alerts for likes, comments, messages
- Optimized database queries for performance, implemented pagination for scalability

- Deployed on local server for testing and demonstration
- Demonstrated full-stack web development expertise from database design to UI/UX

## Goal Sensing Machine: Automated Football Goal Detection

2020

Department of ECE, RUET

Supervisor: Md. Nahiduzzaman | Credit: 0.75

- Developed Arduino-based automated system to detect when football crosses goal line
- Hardware components: Arduino Uno microcontroller, infrared (IR) sensors, laser diodes, photodetectors, buzzer, LED indicators, goal frame prototype
- Sensor configuration: multiple IR beam-break sensors placed across goal line plane
- Detection logic: sequential triggering algorithm to confirm ball passage vs. false triggers
- Implemented debouncing: software filtering to eliminate noise and vibration-induced false positives
- Built user interface: LCD display showing goal count, team scores, match time
- Added audio-visual feedback: buzzer sound, LED flash pattern upon goal detection
- Programmed in Arduino C: interrupt handling, timer functions, sensor polling
- Calibration system: adjustable sensitivity for different ball speeds and sizes
- Tested accuracy: 95%+ detection rate in controlled environment
- Demonstrated skills: sensor integration, real-time embedded programming, sports automation, hardware prototyping
- Potential applications: amateur football leagues, training facilities, recreational centers

## Technical Skills

### Programming Languages:

**Proficient:** Python (5+ years), C (4+ years), C++ (4+ years), MATLAB (3+ years)

**Intermediate:** SQL, PHP, JavaScript, Bash scripting

**Familiar:** HTML5, CSS3, Bootstrap, L<sup>A</sup>T<sub>E</sub>X

### Machine Learning & Deep Learning:

**Frameworks:** TensorFlow, Keras, PyTorch, Scikit-learn, XGBoost, LightGBM

**Techniques:** Supervised learning (classification, regression), Deep neural networks (CNN, RNN, LSTM, Transformers), Transfer learning, Ensemble methods, Semi-supervised learning, Generative models (GANs, VAEs)

**Architectures:** Vision Transformers (ViT), BERT, GPT-2, Vision-Language Models (CLIP, BLIP), EfficientNet, ResNet

**Specialized:** Time-series analysis, Image classification and segmentation, Natural Language Processing, Computer vision, Model interpretability (SHAP, Grad-CAM), Physics-informed machine learning

### Data Science & Analysis:

Pandas, NumPy, SciPy, Matplotlib, Seaborn, Plotly, OpenCV, PIL/Pillow, scikit-image

Feature engineering, Data preprocessing, Dimensionality reduction (PCA, t-SNE, UMAP), Statistical analysis, Data visualization

### Astronomy & Astrophysics Tools:

Astropy, APLpy, Astropy-affiliated packages, Astroquery, FITS file handling, DS9/SAOImage, TOPCAT

Radio astronomy: LOFAR/LoTSS data, VLA/FIRST data, Radio imaging techniques, Radio interferometry

Optical surveys: Pan-STARRS, SDSS, Legacy Survey

Infrared surveys: WISE

Spectroscopy: Hyperspectral data analysis, Spectral unmixing

Coordinate systems: celestial coordinate transformations, cross-matching catalogs, multi-wavelength source matching

### Development Tools & Environments:

Git/GitHub (version control, collaboration), Linux/Unix systems (Ubuntu, command line), Jupyter Notebook/Lab, Google Colab, VS Code, PyCharm

High-Performance Computing (HPC): SLURM job scheduling, parallel computing, cluster computing

Cloud platforms: basic AWS, Google Cloud exposure

### Web Development:

Backend: PHP, Flask (Python), MySQL/PostgreSQL databases

Frontend: HTML5, CSS3, Bootstrap, JavaScript, jQuery

RESTful API design, Database design and normalization

### Hardware & Embedded Systems:

Arduino (C/C++ programming), Raspberry Pi (Python), Microcontroller programming, Sensor integration (IR, ultrasonic, temperature, etc.), IoT systems, Circuit design basics

### Other Technical Skills:

L<sup>A</sup>T<sub>E</sub>X (document preparation, academic writing), Microsoft Office Suite (Word, Excel, PowerPoint), Technical documentation, Research methodology, Scientific writing, Data management



## Certifications & Professional Training

### CASSA Workshop 1: Unclouding Thousand Eyes of Array Radio Telescope

May 2025

Center for Astronomy, Space Science & Astrophysics (CASSA), IUB

🌐 <https://cassa.site/>

- Intensive week-long workshop on radio interferometry and array telescope data processing
- Topics covered: Radio telescope fundamentals, Interferometry principles, LOFAR telescope architecture, Primary beam modeling, Calibration techniques, Imaging pipelines, Data reduction workflows
- Hands-on training: Python programming for radio astronomy, Working with FITS files and astronomical images, Using Astropy and radio astronomy packages, HPC server usage and job submission
- Group project: Developed LOFAR telescope primary beam modeling code using Python
- Implemented beam pattern calculations, Visualized beam response across frequency and sky position, Validated against theoretical models
- Presented findings to workshop participants and instructors
- Gained practical experience with large-scale astronomical data processing on HPC infrastructure
- Networking with radio astronomy researchers and professionals

### Industrial Attachment: Code Studio Software Company

April 2023

Rajshahi, Bangladesh

- One-month intensive industrial training program in machine learning and data science
- Topics covered: Machine learning fundamentals (supervised/unsupervised learning), Time series analysis and forecasting, Feature engineering techniques, Model evaluation and validation, Real-world ML project workflows
- Practical training: Implemented ML algorithms from scratch and using Scikit-learn, Built time series forecasting models (ARIMA, Prophet), Worked on real industry datasets, Learned software development best practices
- Projects: Customer churn prediction model, Sales forecasting using time series, Sentiment analysis of product reviews
- Gained industry perspective on ML application challenges: Data quality issues, Scalability considerations, Production deployment requirements, Client communication and requirement gathering
- Exposure to agile development methodology and team collaboration tools

## Awards & Honors

### International Astronomy & Physics Achievements

- **2017 – Bangladesh National Team Selection:** 11th International Olympiad on Astronomy and Astrophysics (IOAA)  
12-21 November 2017, Phuket, Thailand | 🌐 [ioaa2017.org](https://ioaa2017.org)  
Selected through national olympiad rounds to represent Bangladesh (1 of 5 students); visa complications prevented entire national team from participating
- **2023 – Silver Medal:** The University Physics Competition (International)  
🌐 [uphysicsc.com](https://uphysicsc.com)  
Solved advanced theoretical physics problem over 48-hour period
- **2022 – Accomplished Winner:** The University Physics Competition (International)  
🌐 [uphysicsc.com](https://uphysicsc.com)  
Recognition for outstanding performance in international physics competition
- **2020 – Silver Medal:** The University Physics Competition (International)  
🌐 [uphysicsc.com](https://uphysicsc.com)  
Earned silver medal for solving challenging physics problem
- **2021 – Bronze Honor:** International Astronomy & Astrophysics Competition (IAAC)  
🌐 [iaac.space](https://iaac.space)  
Achieved bronze medal in global astronomy competition
- **2021 – Bronze Honor:** International Youth Math Challenge (IYMC)  
🌐 [iymc.info](https://iymc.info)  
Earned bronze medal in prestigious international mathematics competition
- **2020 – Silver Honor:** International Youth Math Challenge (IYMC)  
🌐 [iymc.info](https://iymc.info)  
Secured silver medal in international mathematics competition

### National Olympiad Achievements

- **2017 – National Winner (Selected for Bangladesh Team):** 12th Bangladesh Astro-Olympiad  
Bangladesh Astronomical Association & Biggan Andolon Moncho
- **2017 – 10th Position (Regional Level):** 12th Bangladesh Astro-Olympiad  
Bangladesh Astronomical Association & Biggan Andolon Moncho

- **2018 – Regional Champion:** 8th Dutch-Bangla Bank-Prothom Alo Bangladesh Physics Olympiad  
Bangladesh Physics Olympiad Committee & Prothom Alo
- **2018 – Honorable Mention (National Round):** 8th Dutch-Bangla Bank-Prothom Alo Bangladesh Physics Olympiad  
Bangladesh Physics Olympiad Committee & Prothom Alo
- **2018 – 8th Position (Regional Level):** 1st Bangladesh Olympiad on Astronomy & Astrophysics  
BDOAA Committee
- **2018 – Participant (National Level):** 1st Bangladesh Olympiad on Astronomy & Astrophysics  
BDOAA Committee
- **2023 – Online Round Winner (Mathematics):** National STEAM Olympiad, Bangladesh  
 [nationalsteamolympiad.com](https://nationalsteamolympiad.com)  
Secured first place in national-level mathematics competition
- **2023 – Top 10 Winner (Physics-RUET):** Precise Energy Olympiad  
Placed in top 10 among university-level participants
- **2022 – Quarter Finalist:** Green Earth Quest  
Advanced to quarter-final round in environmental science and sustainability competition
- **2021 – Honorable Mention:** ICPC Asia Dhaka Regional Site Online Preliminary Contest  
Team: RUET\_Noob\_Coders |  [ICPC Profile: L4CHE9B9411F](#)  
Achieved Honorable Mention in prestigious competitive programming contest

## Other National Science Olympiad Participations (2014–2018)

- **2017:** 7th Dutch-Bangla Bank-Prothom Alo Bangladesh Physics Olympiad (Regional) – Participant
- **2018:** Regional Mathematical Olympiad – Participant, Bangladesh Mathematical Olympiad Committee, Chittagong
- **2017:** Regional Mathematical Olympiad – Participant, Bangladesh Mathematical Olympiad Committee, Chittagong
- **2017:** 1st English Olympiad of Bangladesh – Campus Ambassador & Participant, WOC (World of Champions)
- **2017:** 7th Bangladesh Chemistry Olympiad (Preliminary Round) – Participant, Bangladesh Chemical Society, Chittagong College
- **2017:** Prothom Alo Bhasha Protijog (Regional Round) – Participant, HSBC & Prothom Alo
- **2016:** 6th Bangladesh Physics Olympiad (Divisional Round) – Participant, Bangladesh Physics Olympiad Committee
- **2014:** B.A.S. Divisional Science Olympiad – Participant, Bangladesh Academy of Sciences (BAS), Chittagong Center

## Academic & Scholarship Achievements

- **2020–2022:** Technical Scholarship – University Grants Commission (UGC), Bangladesh  
Three consecutive years, awarded to top-performing students in B.Sc. ECE
- **2016:** SSC Examination – GPA 5.00/5.00, Outstanding Performance Award by Cambrian Education Group
- **2009:** Primary Scholarship Exam (Class 4th) – 2nd Grade, Chittagong Kindergarten Association
- **2008:** Primary Scholarship Exam (Class 3rd) – General Grade, Chittagong Kindergarten Association
- **2007:** Primary Scholarship Exam (Class Two) – General Grade, Chittagong Kindergarten Association
- **2006:** Primary Scholarship Exam (Class One) – First Grade, Chittagong Kindergarten Association

## Leadership & Other Recognition

- **2020 – Role Model Award 2020-2021:** Student Leadership Development Workshops  
US Embassy, Dhaka, Bangladesh  
Selected as Role Model among participants in leadership development program
- **2021 – 1st Place:** Presentation Contest, Einstein's Birthday and Pi Day Celebration 2021  
Astronomy & Science Society of RUET

## Sports Achievements

- **2019 – Player:** ECE'18 Cricket Squad (Long-Pitch Batter) — RUET Inter-Department Fresher's Cricket Tournament
- **2019 – Player:** ECE'18 Football Squad (Goalkeeper) — RUET Inter-Department Fresher's Football Tournament (Quarter-Final)
- **2023 – 3rd Place:** Table Tennis — RUET ECE Inter Series Sports Tournament
- **2023 – Semi-Finalist:** Table Tennis — RUET CESA Table Tennis Tournament
- **2023 – Semi-Finalist:** Carrom (Duo) — RUET CESA Table Tennis Tournament
- **Short Pitch Cricket (All-Rounder) — RUET ECE Inter Series Sports Tournament:** Runner Up (2019), Semi-Final (2022), Semi-Final (2023), Runner Up (2024)
- **Short Pitch Cricket (All-Rounder) — RUET CESA Short Pitch Cricket Tournament:** Runner Up (2019), Semi-Final (2022), Runner Up (2023), Group Stage (2024)
- **Football (Goalkeeper) — RUET ECE Inter Series Sports Tournament:** Semi-Finalist (2019, 2022, 2023, 2024)

- **2020 – Champion:** Football (Goalkeeper) — RUET CESA Football Tournament
- **Long Pitch Cricket — Bangladesh Railway Co-operative Housing Society Shomaj Kollyan Porishod Annual Cricket Tournament:** Champion, All-Rounder (2017); Runner Up, Batter (2018)
- **2016 – Recognition (Shongbordhona):** For achieving GPA 5.00 in SSC — Bangladesh Railway Co-operative Housing Society Shomaj Kollyan Porishod

## Cultural & Extra-Curricular Activities

- **2026 – National Participant:** Teer 16th National Archery Championship 2025 — CTG Archery Club (Recurve Men & Recurve Mixed Team), under Coach Md Jabir Uddin
- **2025 – 5th Position:** 2nd Chittagong Open Archery Championship, CTG Archery Club  
Coach: Md Jabir Uddin (World Archery Level 1 Coach, National Judge, Bangladesh Archery Federation)
- **2025:** Self Defence Training Campaign – Participant, Chittagong Kick-Boxing Wushu Academy (CKBWUA)
- **2009:** Annual Camp – Participant, St. Xavier’s School Cub Scout Group, Chittagong
- **2009:** Cultural Competition (Independence Day) - Drawing – Participant, Maitri Khelaghar Ashar, Pahartali, Chittagong

## Competitive Programming Achievements

- **Total Problems Solved:** 160+ across multiple platforms
- **LeetCode:** [Profile: DebaPriyoGuha](#)  
Solved problems in data structures, algorithms, dynamic programming, graph theory
- **Codeforces:** [Profile: kobeze\\_CDEF\\_solvekorum](#)  
Participated in rated contests; solved problems in competitive programming
- **Codechef:** [Profile: ece\\_1810027](#)  
Regular contest participation; solved algorithmic challenges
- **AtCoder:** [Profile: DebaPriyoGuha](#)  
Participated in Japanese competitive programming platform
- **GitHub:** [github.com/DebaPriyoGuha](https://github.com/DebaPriyoGuha)  
Active open-source contributions; project repositories for research and coursework

## Leadership & Service

### President

*June 2022–May 2023*

*Astronomy & Science Society of RUET (ASSR)*

Rajshahi University of Engineering & Technology

 [assr-hhgt.vercel.app](https://assr-hhgt.vercel.app)

- Led organization of 50+ active members promoting astronomy, astrophysics, and science education
- Organized flagship event: **"AURORA: National Science Festival 2023"**
  - National-level science festival with 500+ participants from universities across Bangladesh
  - Featured competitions: astrophotography, poster presentation, science quiz, science olympiad, project exhibition
  - Invited speakers: renowned astronomers and scientists from national institutions
  - Managed budget, logistics, sponsorships, and event execution
  - Coordinated content creation, graphic design, and marketing teams
  - Served as Team Leader of Science Olympiad Problem Setters’ Team
  - Acted as Judge for Science Olympiad (Mathematics, Physics)
  - Volunteered in event management and coordination
- Conducted regular astronomy observation sessions: telescope nights, meteor shower watching, solar observation
- Organized seminars and workshops: topics in astrophysics, space technology, research methodology
- Organized core programs: Pi Day & Einstein’s Birthday Celebration, Science Talk Series, Astro Talk Sessions, Moon Observation Camps, Star Gazing Nights, Movie Sessions
- Led website development project: modern website showcasing society activities, events, resources
- Mentored junior members: guidance in academic pursuits, research interests, career planning
- Established partnerships with other university science clubs and national organizations
- Managed social media presence: content creation, engagement with astronomy community
- Coordinated participation in national-level science fairs and astronomy olympiads
- Developed leadership, team management, event organization, and communication skills

### School Science Outreach Camps (Resource Person)

*Aug 2022–Sep 2024*

*Astronomy & Science Society of RUET (ASSR)*

- Conducted astronomy and science outreach camps at schools on behalf of ASSR, promoting astronomy and scientific thinking among school students:

- BCCIR Laboratory School (4 August 2022)
- Shahid Police Super Najmul Haque Police Lines School, Joypurhat (13 July 2023)
- Subil High School, Bogura (19 September 2024)

### Research & Planning Secretary

June 2021–May 2022

*Astronomy & Science Society of RUET (ASSR)*

Rajshahi University of Engineering & Technology

- Coordinated research activities and academic programs for society members
- Planned and organized scientific seminars, workshops, and observation sessions
- Developed strategic plans for society growth and member engagement
- Assisted in organizing astronomy competitions and science festivals
- Facilitated collaboration with faculty advisors and external organizations
- Managed documentation and archiving of society activities
- Contributed to development of educational content and resources

### Vice President (Sports)

Aug 2023–Jan 2024

*Chittagong Engineering Students' Association (CESA)*

RUET Chapter, Rajshahi

- Led sports activities for students from Chattogram studying at RUET
- Organized and managed cricket and football teams: player recruitment, practice schedules, team selection
- Coordinated inter-department and inter-university sports tournaments
- Arranged friendly matches with other student associations
- Managed sports equipment, ground bookings, and logistics
- Fostered teamwork, competitive spirit, and physical fitness among members
- Organized social events and networking opportunities for Chattogram students
- Developed skills in team leadership, strategy planning, and sports management
- Received Certificate of Recognition for excellence and outstanding leading quality

### Assistant Information & Publication Secretary

Sept 2022–Aug 2023

*Chittagong Engineering Students' Association (CESA)*

RUET Chapter, Rajshahi

- Managed information dissemination and publication activities for CESA
- Coordinated social media presence and digital content creation
- Designed promotional materials and event posters
- Maintained communication channels with members and stakeholders
- Assisted in documentation of association activities and events
- Developed skills in graphic design, content writing, and digital marketing

### Academic Team Member

2019–Present

*Bangladesh Mathematical Olympiad (BdMO)*

*Bangladesh Physics Olympiad (BdPhO)*

*Bangladesh Olympiad on Astronomy & Astrophysics (BDOAA)*

- Volunteered as academic team member for national-level olympiad competitions
- **Question Creation:** Designed challenging mathematics, physics, and astronomy problems for regional and national rounds
- **Answer Checking:** Evaluated student solutions ensuring fair and consistent grading
- Contributed to problem vetting process: reviewing difficulty, clarity, originality
- Participated in training camps: mentored students preparing for international olympiads
- Helped identify and nurture mathematical and scientific talent among Bangladeshi students
- Collaborated with university professors and olympiad alumni
- Gained experience in problem-solving pedagogy and educational assessment
- Stayed connected with competitive mathematics and physics community

### Financial Secretary (In-charge)

2017–2019

*Chittagong Math Circle*

Chattogram, Bangladesh

- Managed financial operations for mathematics enrichment organization
- Handled budget planning, expense tracking, and financial reporting
- Coordinated fundraising activities and sponsorship management
- Maintained transparent financial records and accountability
- Supported organization of mathematics workshops, problem-solving sessions, and olympiad training
- Assisted in outreach programs to promote mathematical literacy among students
- Developed skills in financial management and organizational administration

### Executive Member

2018–2020

*Paradox Physics School*

Chattogram, Bangladesh

- Contributed to physics education and olympiad training programs
- Assisted in organizing physics workshops and problem-solving sessions
- Mentored junior students preparing for physics olympiads
- Participated in curriculum development for physics enrichment courses
- Helped coordinate events, seminars, and guest lectures
- Promoted physics education and scientific thinking among high school students

#### Founder & Lead Instructor

*Dec 2018–Present*

*STEM by Priyojeet*

Private Tutoring Initiative, Chattogram

- Founded and operated private tutoring service specializing in STEM education
- **Students Mentored: 18+** across various educational levels
- **Subject Areas:** Physics (mechanics, electromagnetism, thermodynamics, modern physics), Mathematics (calculus, algebra, geometry, number theory, problem-solving), Programming (Python, C, algorithms), Other Science subjects (chemistry, biology - occasionally)
- **Teaching Methodology:** Conceptual understanding over rote memorization, Problem-solving strategies and critical thinking, Hands-on demonstrations and real-world applications, Olympiad-style training for advanced students, Personalized curriculum based on individual student needs
- **Student Achievements:**
  - Multiple students gained admission to top colleges in Bangladesh
  - Students secured places in medical colleges (MBBS programs)
  - Students admitted to prestigious engineering universities (BUET, RUET, CUET)
  - Students gained admission to NDC and other top-tier institutions
  - One exceptional student: completing university-level coursework while only in Class 9, demonstrating advanced mathematical and programming capabilities
- Developed custom study materials, problem sets, and practice exams
- Provided career counseling and academic guidance beyond classroom instruction
- Built long-term mentoring relationships extending beyond formal tutoring
- Adapted teaching methods during COVID-19: transitioned to online platforms, created digital content
- Maintained high student satisfaction and parent communication
- Demonstrated passion for education and knowledge dissemination
- Honed communication, pedagogy, and mentorship skills

## Professional Development & Activities

#### AI Training Contributor

*2024–Present*

*Outlier AI*

Remote/Online Position

- Contributing to training and evaluation of Large Language Models (LLMs) for AI companies
- **Domain Expertise:** Mathematics problem-solving, Programming and algorithms, Code generation and debugging
- Creating high-quality training data: writing mathematical problems and solutions, coding challenges and implementations, detailed explanations and step-by-step reasoning
- Evaluating model outputs: assessing correctness, clarity, completeness of AI-generated responses, identifying errors and suggesting improvements
- Providing human feedback to improve model performance in STEM domains
- Rating model responses across multiple dimensions: accuracy, reasoning quality, code efficiency
- Participating in specialized projects requiring domain expertise
- Collaborating with global team of subject matter experts
- Gaining insight into cutting-edge LLM training methodologies and AI development
- Contributing to advancement of AI capabilities in technical domains
- Earning supplementary income while developing expertise in AI evaluation

#### Competitive Archer

*2025–Present*

*CTG Archery Club*

Chattogram, Bangladesh

- Active member and competitive participant in archery club
- **Achievement: 5th Position** – 2nd Chittagong Open Archery Championship
- Regular training sessions: recurve bow technique, target shooting, distance shooting
- Participated and represented the club in Recurve Men and Recurve Mixed Team events at the Teer 16th National Archery Championship 2025
- Participated in regional and national archery competitions

- **Archery Trainer** (January 2026–Present): training newcomers and assisting national preparation; precision and accuracy specialist for national-level preparations
- Developed mental focus, hand-eye coordination, and discipline
- Balancing athletic pursuit with academic and research commitments
- Applying principles of precision and consistency from archery to research work

## Languages

**Bengali:** Native speaker; full professional proficiency in reading, writing, speaking

**English:** Fluent; advanced academic and professional proficiency

- IELTS Academic: Overall 7.0 (Listening: 8.0, Reading: 7.5, Writing: 6.5, Speaking: 6.0)
- Test Date: 22 September 2025
- Extensive experience in technical writing, research paper composition, academic presentations
- Regular consumption of English scientific literature, lectures, and media

**Hindi/Urdu:** Advanced proficiency in listening and speaking; passive understanding through media consumption.

- Can understand Bollywood movies, news, and conversations without subtitles
- Comfortable speaking in informal contexts
- Limited formal writing ability

## Personal Interests & Hobbies

### Martial Arts:

- Practicing martial arts for physical fitness, discipline, and self-defense
- Completed Self Defence Training Campaign at Chittagong Kick-Boxing Wushu Academy
- Interest in various martial arts forms and philosophies
- Applying principles of focus and perseverance to academic work

### Whistling Melodies:

- Skilled at whistling complex musical melodies and compositions
- Repertoire includes classical music, film songs, and contemporary pieces
- Creative outlet for stress relief and artistic expression
- Entertaining friends and bringing joy through musical whistling

### Late Night Street Exploration:

- Enjoying peaceful late-night walks through city streets
- Observing urban landscapes, architecture, and nocturnal atmosphere
- Time for reflection, thinking, and mental relaxation
- Exploring different neighborhoods and discovering hidden spots
- Finding inspiration and clearing mind for creative problem-solving

### Other Interests:

- Reading science fiction and popular science books
- Following developments in space exploration and astronomy news
- Photography, especially astrophotography and night sky imaging
- Participating in online science and astronomy communities
- Watching scientific documentaries and educational content
- Playing strategy-based video games and chess

## References

### Khan Muhammad Bin Asad, PhD

Founder & Director, Center for Astronomy, Space Science & Astrophysics (CASSA)

Assistant Professor, Department of Physical Sciences

School of Engineering, Technology and Sciences

Independent University, Bangladesh (IUB)

✉ [kasad@iub.edu.bd](mailto:kasad@iub.edu.bd)

☎ +880-1320-767186

🌐 <https://cassa.site/people/asad/>

*Relationship:* Current research supervisor at CASSA; Principal Investigator for RGC v2.0 project and Vision-LLM research

### Md. Azmain Yakın Srizon

Assistant Professor

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☎ +880-1790-187189  
🌐 <https://www.ruet.ac.bd/azmainsrizon>

*Relationship:* Research supervisor at Young Learners' Research Lab; mentor for LunaXNet and multiple research projects; Advisor of Astronomy & Science Society of RUET

### **Fariya Tabassum**

Assistant Professor & Former Head  
Department of Electrical & Computer Engineering  
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🌐 <https://www.ruet.ac.bd/fariya>

*Relationship:* Undergraduate thesis supervisor (PIE-ADA project); Department Head during final year

### **Md. Nazmul Sharif**

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Department of Mathematics  
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🌐 <https://www.ruet.ac.bd/nazmul06024160>

*Relationship:* Academic mentor; collaborated on olympiad activities; former Advisor of Astronomy & Science Society of RUET; knows student through mathematics courses and competitions

*Last Updated: June 2026*

*Full CV available upon request*